

*MA.7.NSO.2.2***Benchmark**

**MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.**

**Connecting Benchmarks/Horizontal Alignment**

- MA.7.AR.1.1
- MA.7.AR.2.1, MA.7.AR.2.2
- MA.7.AR.3.1, MA.7.AR.3.2, MA.7.AR.3.3
- MA.7.AR.4.1, MA.7.AR.4.2, MA.7.AR.4.3, MA.7.AR.4.4, MA.7.AR.4.5
- MA.7.GR.1.1, MA.7.GR.1.2, MA.7.GR.1.3, MA.7.GR.1.4, MA.7.GR.1.5
- MA.7.GR.2.1, MA.7.GR.2.2, MA.7.GR.2.3
- MA.7.DP.1.1, MA.7.DP.1.2, MA.7.DP.1.3
- MA.7.DP.2.1

**Terms from the K-12 Glossary**

- Rational Number

**Vertical Alignment****Previous Benchmarks**

- MA.6.NSO.2.1, MA.6.NSO.2.2
- MA.6.NSO.4.1, MA.6.NSO.4.2

**Next Benchmarks**

- MA.8.NSO.1.5, MA.8.NSO.1.7

**Purpose and Instructional Strategies**

In grade 6, students performed operations with integers, multiplied and divided positive multi-digit numbers with decimals to the thousandths and computed products and quotients of positive fractions by positive fractions, including mixed numbers with procedural fluency. In grade 7, students perform all four operations with positive and negative rational numbers with procedural fluency. In grade 8, they will expand to operations with rational numbers including exponents and radicals and will perform operations with rational numbers expressed in scientific notation.

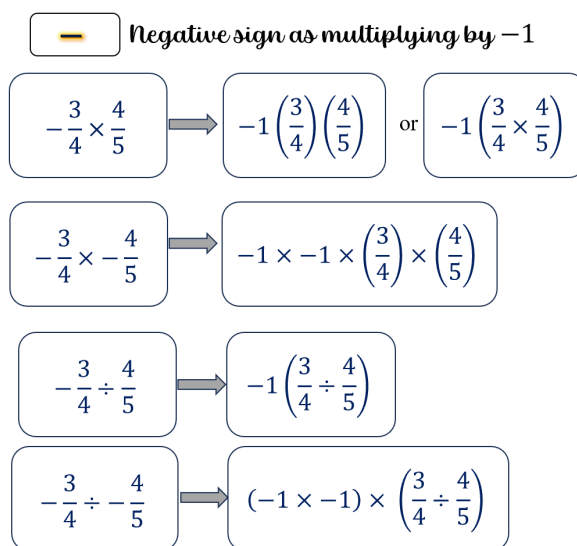
- This benchmark is the completion of arithmetic operations with rational numbers (*MTR.3.1*).
- Instruction includes the possibility that the division of two fractions can be written as a complex fraction. This connection will be important when students work with algebraic expressions in later grades.
- Students should develop fluency with and without the use of a calculator when performing operations with rational numbers.

**Common Misconceptions or Errors**

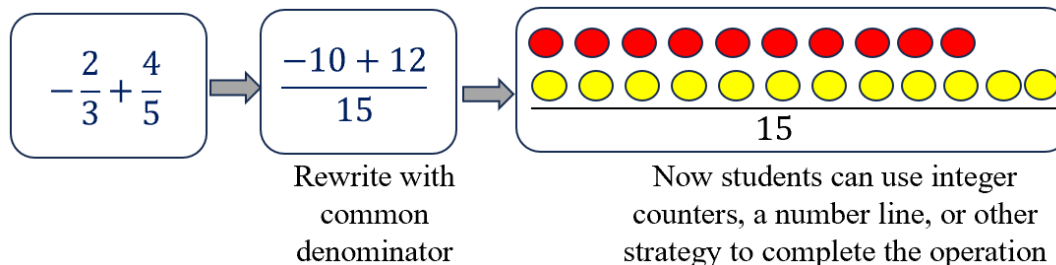
- Students may think the product of a fraction and another fraction is greater than either factor. Use manipulatives or models referenced in previous grade levels to support conceptual understanding (*MTR.2.1*).
- Students may incorrectly believe that dividing by  $\frac{1}{2}$  is the same as dividing by 2.
- Students may incorrectly solve complex fractions by multiplying the two fractions.

### Strategies to Support Tiered Instruction

- Instruction includes the use of fraction tiles to represent operations with positive fractions while simultaneously recording the equivalent numerical expressions.
- Instruction includes the use of base ten blocks to represent operations with positive decimals while simultaneously recording the equivalent numerical expressions.
- Instruction includes the use of two-color counters to represent operations with integers while simultaneously recording the equivalent numerical expressions.
- Teacher co-creates a graphic organizer with students to review operations with positive fractions and operations with integers to assist when applying operations with rational numbers.



- Instruction includes using manipulatives or models referenced in previous grade levels to support conceptual understanding.



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**Instructional Tasks**

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*Instructional Task 1 (MTR.7.1)*

Daliah purchases eggs by the dozen for her two children. Each day, Zane eats  $\frac{1}{4}$  carton and Amare eats  $\frac{1}{6}$  carton. A carton of 12 eggs costs \$1.65.

Part A. How much does Daliah spend on eggs for her two children in 30 days?

Part B. During one of her shopping trips, Daliah finds that her grocery store has started to sell cartons of 18 eggs for \$2.25. If she begins to purchase these cartons, how much does Daliah spend on eggs for her two children in 30 days? After how many days will Daliah spend more than \$50? Explain your reasoning.

*Instructional Task 2 (MTR.3.1)*

Given  $a = -2\frac{3}{5}$  and  $b = \frac{2}{3}$ , calculate the following:

- $a + b$
- $a - b$
- $a \cdot b$
- $\frac{a}{b}$

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**Instructional Items**

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*Instructional Item 1*

Determine the product of  $\frac{15}{6}$  and  $-1.2$ .

*Instructional Item 2*

What is the value of the expression  $7.24 - 5.01 - 78.4$ ?

*Instructional Item 3*

What is the value of the expression  $\frac{\frac{24}{5}}{\frac{1}{9}}$ ?

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*\*The strategies, tasks and items included in the BIG-M are examples and should not be considered*